

trust in, we have two problems: how do we agree on a common reality, and how do we make our economy robust (that is to say, how do we prevent one from fooling many into believing a false reality)?

The first problem has a simple solution. If everyone agrees on the history of transactions, any disagreement or false claims can be settled by majority. So we broadcast transactions publicly and store their order in a common database, or ledger. The ledger itself is the blockchain. Everyone has a copy of it, and everyone is continuously updating it. Whose update gets written to history depends on another concept which can be 'proof of work' or 'proof of stake'. This is also the solution to the second problem, as the proof decides reality. Technically we democratizing trust and not eliminating the need for it. Still, pretty neat, isn't it? The best part is, it is not limited to a list of transactions, the actual blocks in the blockchain could be about all sorts of data! But enough talk about ideas, now we shall discuss implementations!

Immutable history

So a consensual history sounds fancy, almost utopian, but how do we do that? This is where blockchain comes in, very literally. If you understand this, you understand blockchain. But before that, we must do a quick review of a couple of cryptographic concepts - Hashes and Merkle Trees.

A hash is an output of a special function, called a cryptographic hash function (surprise!) such as SHA2, which has the property that the output is of fixed length for any given length of input. The output is usually much quicker to compute from the input. This property is useful for proof of work/stake applications. Ideally, every unique input will give a unique output, which is why it is colloquially called the fingerprint of the data that is being hashed. If we change even one character in a million character input to the hash function, we get a very different output which lets us know that the data has been corrupted/tampered with.

The Merkle tree is a data structure where the leaf nodes are labelled with the hash digests of some data, and every other node is labelled with the hash digest of its child nodes. This cascaded hashing allows for storing reduced amounts of hash digest values for a large set of data. Since we are essentially grouping hash digests into a single hash layer by layer, the older unchanging data can be discarded. Version control software such as Git use this structure for tracking changes and verifying integrity.

When it comes to Bitcoin, the ledger of transactions are themselves the 'coin'. A particular number of transactions compose a 'block' and this data